■■ RepoGuardian — Team & AI Setup Guide

Hackathon PS-04 • Multi-Branch Collaboration & Environment Setup

1. Executive Overview & Repository Uplink

This guide provides exact, step-by-step instructions for each team member and their respective AI coding assistants (Claude, Gemini, Cursor, Copilot) to set up, build, and extend **RepoGuardian** without merge conflicts or code regressions.

Parameter	Value / URL
GitHub Remote	https://github.com/richa866/RepoGuardiann.git
Primary Branch	main (Production / Demo Ready)
Backend Stack	Python 3.10+, FastAPI, SQLite, ChromaDB, Sentence-Transformers
Frontend Stack	React 19, Vite, @react-three/fiber, Three.js, Tailwind CSS, Recharts
Default Ports	Backend: 8000 | Frontend: 5173

2. Quick Start: Clone & Local Setup

Run these commands in terminal to set up the full working platform locally:

```
# 1. Clone the repository
git clone https://github.com/richa866/RepoGuardiann.git
cd RepoGuardiann

# 2. Setup & Start Backend (Terminal 1)
cd backend
python3 -m venv venv
source venv/bin/activate # On Windows: venv\Scripts\activate
pip install -r requirements.txt
python seed_dummy_data.py
uvicorn app.main:app --reload --port 8000

# 3. Setup & Start Frontend (Terminal 2)
cd ../frontend
npm install
npm run dev
```

3. Team Roles & Dedicated Branch Ownership

Each team member (and their AI assistant) should switch to their dedicated branch before writing code:

Role & Branch	Assigned Files	Key Responsibilities & AI Guardrails
Backend Dev feat/backend-pipeline	backend/app/sync.py backend/app/monitor.py backend/app/github_client.py	• Manage GitHub API pagination and rate limits. • Ensure continuous monitoring loop enqueues subtasks. • Maintain /health-metrics and /monitor/status.

AIML #1 (RAG Lead) feat/rag-retrieval	backend/app/rag.py backend/app/tools.py	• Tune Chroma embedding functions (all-MiniLM-L6-v2). • Ensure maintainer resolution context (e.g. *'Fixed in v2.1'*) is embedded alongside text. • Evaluate similarity search accuracy on real repos.
AIML #2 (Agent Lead) feat/agent-escalation	backend/app/agent.py backend/app/tools.py backend/app/llm.py	• Refine 6 independent tool functions (duplicate, SLA, security, stale, missing info, contentious). • Calibrate synthesis prompts with evidence citations (similarity %, days unresponded). • Verify human override feedback flow.
AIML #3 / Frontend feat/frontend-hud	frontend/src/components/3d/ frontend/src/components/hud/ frontend/src/App.jsx	• Polish 3D Git tree & clustered issue scene. • Enhance glassmorphism HUD, feedback buttons, and charts. • Test live repo connect modal (httpie/cli, psf/black).

4. Specific Instructions for Teammates' AI Assistants

When prompting your AI assistant (e.g., in Cursor, Antigravity, Claude, or Copilot), provide the following guidelines:

- **Documentation & Arch Integrity:** Do not rewrite or delete existing working endpoints or 3D components unless specifically instructed.
- **Dual Mode Resiliency (Gemini & Fallback):** The backend automatically supports both Google Gemini (GEMINI_API_KEY) and deterministic rule-based synthesis fallback. Never remove the fallback code path.
- **Multi-Step Tool Paradigm:** Maintain the architecture where evaluate_issue() executes 6 independent tool functions before calling LLM synthesis. Do not condense tool checks into a single generic prompt.
- **3D Asset Location:** All Blender 3D models reside in frontend/public/models/ (repoguardian_logo.glb, git_branch_node.glb, smooth_issue_orb.glb, commit_node.glb).
- **Local Vector Store:** Chroma DB is local and persistent under backend/data/chroma. No external vector cloud API is required.

5. Environment Configuration (.env Template)

To connect real live GitHub repositories and Gemini API, create backend/.env:

```
# Optional: Personal Access Token (raises rate limit to 5000/hr)
GITHUB_TOKEN=ghp_your_personal_access_token_here

# Optional: Auto-connect repository on boot (e.g. httpie/cli, psf/black, pallets/flask)
GITHUB_REPO=httpie/cli

# Optional: Gemini API Key for LLM prose synthesis (free from aistudio.google.com)
GEMINI_API_KEY=AIzaSy...
GEMINI_MODEL=gemini-1.5-flash

# Background Scheduler Interval (seconds)
MONITOR_POLL_INTERVAL_SECONDS=300

# Storage Paths
DATABASE_PATH=./data/repoguardian.db
CHROMA_PATH=./data/chroma
```

6. Git Workflow & Merging Process

Follow these rules to prevent merge collisions and maintain code quality:

1. **Checkout your branch:** git checkout feat/your-feature-name
2. **Stay updated with main:** Regularly run git fetch origin && git merge origin/main
3. **Test before pushing:** Run npm run build in frontend and verify backend responds on /health.
4. **Push to remote:** git push origin feat/your-feature-name
5. **Merge to main:** Open a Pull Request on GitHub or fast-forward merge once verified.

7. Verified Demo Repositories Cheat Sheet

These public repositories are verified and pre-tested for live demoing:

Repository	Key Demo Highlights	Flagged Issue Numbers
httpie/cli	Rich security vulnerabilities, plaintext auth credentials, terminal escape injection.	• #1812 (Escape injection) • #1710 (Plaintext credentials) • #1636 (Stale untriaged)
psf/black	Well-labeled, regression-heavy, supply-chain security proposals.	• #5029 (Zip Slip security) • #3665 (Duplicate bug) • #5120 (Stale immutable releases)
pallets/flask	Mature historical decisions ('closed as duplicate of #X', 'fixed in v2.1'). Ideal for RAG.	• #6114 (Duplicate resolution) • #6044 (Teardown bug)